

Effect of Vehicle Speed on Wireless Communication: Doppler Shift and Rayleigh Fading Analysis

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1 Objective

To investigate the effects of changing vehicle or the receiver speed affects the Wireless communication quality by studying the impact of Doppler shift and Rayleigh fading on the Bit Error Rate (BER) of a Quadrature Phase Shift Keying (QPSK) communication systems.

2 Problem Statement

Mobile Wireless communication systems must contend with a time-varying channel caused by the relative motion between the transmitter and receiver. This motion introduces a Doppler shift in the received signal frequency, and when combined with multipath propagation, gives rise to Rayleigh fading, a rapid and random fluctuations in the received signal envelope. As vehicle speed increases, the channel decorrelates more quickly (shorter coherence time), which can degrade the reliability of digital communication systems that rely on accurate channel knowledge at the receiver.

This project investigates the extent to which vehicle speed affects wireless link performance, distinguishing between the fundamental statistical effect of fading itself and the practical effect of a receiver's ability (or inability) to track a fast-changing channel using periodic pilot symbols.

3 Theory and Background

3.1 Doppler Fading

Doppler Fading is a unique aspect of wireless communication systems. The Doppler shift is fundamental principle related to the electromagnetic radio-wave propagation. The Doppler shift associated with an electromagnetic wave is defined as the perceived change in the frequency of the wave due to relative motion between the transmitter and receiver.

The Doppler shift is given by [?]:

$$f_d = \frac{v}{\lambda} \cos \theta = \frac{v f_c}{c} \cos \theta \quad (1)$$

where θ is the angle of the line joining the mobile receiver and the base station, v is the velocity of the mobile receiver, f_c is the carrier frequency, and c is the speed of light.

3.2 Multipath Propagation and Rayleigh Fading

In a mobile environment without a dominant line-of-sight path, the received signal is the sum of many multipath components arriving from different angles, each with a random amplitude, phase, and Doppler shift. By the Central Limit Theorem, the in-phase and quadrature components of the resulting received signal approach independent, zero-mean Gaussian random processes for a large number of scatterers [?]. The envelope of this process, $|h(t)|$, is Rayleigh distributed:

$$p(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r \geq 0 \quad (2)$$

This model, known as *Clarke's reference model*, forms the theoretical basis for simulating Rayleigh fading channels.

3.3 Jakes Doppler Power Spectrum

The multipath components do not arrive with a single Doppler shift, but rather span the range $[-f_{d,max}, f_{d,max}]$, where $f_{d,max}$ is the maximum frequency deviation. The resulting Doppler power spectral density, known as the Jakes spectrum, is given by:

$$S(f) = \frac{1}{\pi f_{d,max} \sqrt{1 - (f/f_{d,max})^2}}, \quad |f| < f_{d,max} \quad (3)$$

This characteristic “bathtub” shape governs how quickly the fading channel varies in time: a larger $f_{d,max}$ (higher speed) corresponds to a wider spectral spread, and therefore a faster-varying channel envelope.

3.4 Coherence Time

The coherence time T_c is a statistical measure of the time duration over which the channel can be considered approximately constant. A common approximation is [?]:

$$T_c \approx \frac{9}{16\pi f_{d,max}} \quad (4)$$

As speed increases, $f_{d,max}$ increases, and T_c which means that the channel changes more rapidly, and any channel knowledge obtained by the receiver becomes outdated more quickly.

3.5 Rayleigh Fading Channel Simulation Methods

Two independent methods were used in this project to generate a discrete-time, correlated Rayleigh fading channel $h(t)$:

3.5.1 Smith's Spectral Simulation Method

This method shapes complex white Gaussian noise in the frequency domain by the square root of the Jakes power spectrum (Eq. 3), and then applies an inverse FFT to obtain a time-domain fading waveform. This approach is a discrete, computationally efficient realization of the filtering technique originally proposed by Jakes [?] and formalized by Smith.

3.5.2 Zheng–Xiao Sum-of-Sinusoids Method

This method constructs $h(t)$ directly as a sum of a small number (M) of sinusoids with randomized phase, angle of arrival, and initial phase parameters [?]:

$$X_c(t) = \frac{2}{\sqrt{M}} \sum_{n=1}^M \cos(\psi_n) \cos(w_d t \cos \alpha_n + \phi) \quad (5)$$

$$X_s(t) = \frac{2}{\sqrt{M}} \sum_{n=1}^M \sin(\psi_n) \cos(w_d t \cos \alpha_n + \phi) \quad (6)$$

with $\alpha_n = \frac{2\pi n - \pi + \theta}{4M}$, and θ , ϕ , ψ_n independently and uniformly distributed on $[-\pi, \pi]$. Zheng and Xiao proved that this construction yields exact second-order statistics (autocorrelation) matching the theoretical Rayleigh reference model, even for small M (e.g., $M = 8$), unlike the original deterministic Jakes simulator.

3.6 QPSK Modulation

Quadrature Phase Shift Keying (QPSK) maps pairs of bits to one of four constellation points on the unit circle. With Gray-coded mapping and normalized average symbol energy $E_s = 1$, the transmitted symbol is:

$$s = \frac{1}{\sqrt{2}} [(2b_1 - 1) + j(2b_0 - 1)] \quad (7)$$

where $b_1, b_0 \in \{0, 1\}$ are the two bits mapped to that symbol.

3.7 Channel Model and Signal-to-Noise Ratio

The received signal after passing through a flat-fading channel with additive white Gaussian noise (AWGN) is modeled as:

$$y(t) = h(t) s(t) + n(t) \quad (8)$$

Since QPSK conveys 2 bits per symbol, $E_s = 2E_b$. With $E_s = 1$, the noise spectral density corresponding to a given E_b/N_0 (linear) is:

$$N_0 = \frac{1}{2(E_b/N_0)} \quad (9)$$

3.8 Equalization and Channel State Information (CSI)

To recover the transmitted symbol, the receiver must undo the channel's effect, which requires knowledge of $h(t)$ – referred to as Channel State Information (CSI). Two CSI scenarios were studied:

- **Perfect CSI:** the receiver has exact, instantaneous knowledge of $h(t)$.
- **Estimated CSI:** the receiver estimates $h(t)$ only at known pilot symbol positions and interpolates between them, as is done in practical systems.

Two equalization strategies were implemented:

$$\text{Zero-Forcing (ZF): } \hat{s} = \frac{y}{h} \quad (10)$$

$$\text{MMSE: } \hat{s} = \frac{h^*}{|h|^2 + N_0} y \quad (11)$$

ZF perfectly inverts the channel but amplifies noise significantly when $|h|$ is small (deep fade). MMSE balances channel inversion against noise amplification, and reduces to ZF at high SNR.

3.9 Theoretical QPSK BER over Rayleigh Fading

For QPSK with coherent detection over a Rayleigh fading channel under perfect CSI, the average BER as a function of average E_b/N_0 (γ) has the closed-form expression [?]:

$$P_b = \frac{1}{2} \left(1 - \sqrt{\frac{\gamma}{1 + \gamma}} \right) \quad (12)$$

This expression is used throughout this report as the reference for validating simulation results.